

Learning Outcomes

Knowledge and Understanding

KU1: Grasp the relation between source code, the interpreter, and the machine.

KU2: Choose appropriate data types and data structures for different kinds of data, depending on their performance characteristics.

KU3: Design algorithms to solve simple programming problems.

Competence and Skills

CS1: Structure small programs by the use of concepts such as iterations, functions, modules, classes, and methods.

CS2: Structure larger programs into manageable and reusable units.

CS3: Form readable, descriptive and well-documented program code.

CS4: Use programming for basic data analysis involving large textual or numeric files.

CS5: Express mathematical formulas as programming language expressions and algorithms.

CS6: Build basic interactive programs with text-based (and graphical) user interfaces.

CS7: Test programs, for instance using unit testing.

CS8: Use programming tools such as text editor, command line interface, and IDE (integrated development environment).

CS9: Use standard libraries and follow best programming practices.

CS10: Apply basic tools and methods supporting an inclusive cooperation in group work, including JML aspects.

Judgement and Approach

JA1: Assess the difficulty and resources needed for typical programming tasks.